Chapter 6 — Heat and Thermodynamics

Board-style MCQs · PECTAA Physics 11 · Answer key at end

Practice MCQs

1. The ideal gas equation is:

(a) PV = constant always **(b)** P/T = constant always **(c)** PV = nRT **(d)** V/T = constant always

2. The Boltzmann constant is:

(a) 1.38×10^{23} J K^{-1} **(b)** 1.38×10^{-23} J K^{-1} **(c)** 6.02×10^{23} J K^{-1} **(d)** 8.31 J K^{-1}

3. The general gas constant R is about:

(a) 8.31 J mol^{-1} K^{-1} **(b)** 1.38×10^{-23} J K^{-1} **(c)** 6.02×10^{23} J mol^{-1} **(d)** 9.8 J mol^{-1} K^{-1}

4. A real gas behaves ideally at:

(a) low temperature only **(b)** high pressure and low temperature **(c)** low pressure and high temperature **(d)** high pressure only

5. Boyle's law states at constant T:

(a) $P \propto V$ **(b)** $P \propto T$ **(c)** $V \propto T$ **(d)** $P \propto 1/V$

6. Charles' law states at constant P:

(a) $P \propto 1/V$ **(b)** $P \propto T$ **(c)** $V \propto T$ **(d)** $V \propto 1/T$

7. Gay-Lussac's law states at constant V:

(a) $V \propto T$ **(b)** $V \propto 1/T$ **(c)** $P \propto T$ **(d)** $P \propto 1/V$

8. Internal energy of an ideal gas depends only on:

(a) temperature **(b)** container shape **(c)** volume **(d)** pressure

9. The first law of thermodynamics is:

(a) $Q = \Delta U + W$ **(b)** $W = Q + \Delta U$ **(c)** $Q = \Delta U - W$ always **(d)** $\Delta U = QW$

10. Work done by a gas at constant pressure is:

(a) $V\Delta P$ **(b)** P/V **(c)** $\Delta P/\Delta V$ **(d)** $P\Delta V$

11. The area under a P–V graph gives:

(a) heat only **(b)** entropy **(c)** work done **(d)** internal energy only

12. In an isothermal process of an ideal gas:

(a) W = 0 always **(b)** $\Delta U = 0$ **(c)** P = 0 **(d)** Q = 0

13. In an adiabatic process:

(a) Q = 0 **(b)** P constant **(c)** $\Delta U = 0$ **(d)** T constant

14. For an adiabatic process:

(a) PV^γ = constant **(b)** P/T = constant **(c)** V/T = constant **(d)** PV = constant

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15. In adiabatic expansion, temperature:

(a) remains same **(b)** becomes zero **(c)** falls **(d)** rises

16. An isothermal process must be:

(a) slow **(b)** explosive **(c)** very fast **(d)** insulated only

17. An adiabatic curve on P–V graph is:

(a) horizontal **(b)** steeper than isothermal **(c)** flatter than isothermal **(d)** same as isothermal

18. A heat engine converts:

(a) heat into work **(b)** charge into mass **(c)** mass into charge **(d)** work into heat only

19. Efficiency of a heat engine is:

(a) Q_1/Q_2 **(b)** Q_1/W **(c)** W/Q_1 **(d)** Q_2/Q_1

20. Carnot efficiency is:

(a) $1 - T_1/T_2$ **(b)** T_1/T_2 **(c)** T_2/T_1 **(d)** $1 - T_2/T_1$

21. No engine can be more efficient than:

(a) diesel engine **(b)** steam engine **(c)** petrol engine **(d)** Carnot engine

22. A heat engine cannot be 100% efficient because:

(a) friction is zero **(b)** fuel is expensive **(c)** work is zero **(d)** some heat must be rejected

23. A refrigerator is:

(a) a heat engine forward **(b)** a Carnot engine only **(c)** a heat engine in reverse **(d)** a perpetual machine

24. Coefficient of performance of refrigerator is:

(a) W/Q_2 **(b)** Q_1/Q_2 **(c)** Q_2/W **(d)** Q_1/W

25. A refrigerator does not violate the second law because:

(a) external work is supplied **(b)** it is 100% efficient **(c)** no heat is rejected **(d)** it needs no compressor

26. Entropy change is:

(a) $\Delta Q/T$ **(b)** $\Delta Q + T$ **(c)** $\Delta Q \times T$ **(d)** $T/\Delta Q$

27. In a natural irreversible process, total entropy:

(a) increases **(b)** is zero always **(c)** decreases **(d)** is infinite

28. Entropy is a measure of:

(a) disorder **(b)** pressure only **(c)** volume only **(d)** temperature only

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29. Heat death of universe refers to:

(a) maximum entropy and no available work **(b)** zero energy **(c)** maximum pressure **(d)** zero temperature everywhere

30. When ice melts at 0°C, entropy:

(a) decreases **(b)** becomes zero **(c)** increases **(d)** remains same

31. Thermal equilibrium means:

(a) zero temperature **(b)** zero pressure **(c)** no energy **(d)** no net heat flow

32. In a cyclic process, change in internal energy is:

(a) Q **(b)** W **(c)** zero **(d)** maximum

33. Work done on a gas is:

(a) always zero **(b)** equal to Q always **(c)** always positive **(d)** negative if by-system convention

34. The kinetic theory assumes collisions are:

(a) always sticky **(b)** perfectly elastic **(c)** absent **(d)** perfectly inelastic

35. Pressure of a gas is due to:

(a) gravity only **(b)** molecular collisions with walls **(c)** container colour **(d)** molecular weight only

36. Average kinetic energy of gas molecules is proportional to:

(a) volume only **(b)** absolute temperature **(c)** pressure only **(d)** density only

37. At constant volume, pressure of a gas increases when:

(a) temperature decreases **(b)** volume increases **(c)** temperature increases **(d)** moles decrease

38. At constant pressure, volume of a gas increases when:

(a) temperature decreases **(b)** temperature increases **(c)** pressure increases **(d)** moles decrease

39. At constant temperature, pressure increases when:

(a) volume decreases **(b)** temperature decreases **(c)** moles decrease **(d)** volume increases

40. A reversible process is:

(a) always adiabatic **(b)** infinitely slow and retraceable **(c)** always fast **(d)** explosive

41. Free expansion of a gas is:

(a) reversible **(b)** adiabatic always only **(c)** irreversible **(d)** isothermal always

42. The second law forbids:

(a) complete conversion of heat to work in a cycle **(b)** any heat engine **(c)** any work **(d)** any refrigerator

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43. In a Carnot cycle, efficiency depends only on:

(a) engine size **(b)** reservoir temperatures **(c)** fuel type **(d)** working substance

44. If $T_1 = 600$ K and $T_2 = 300$ K, Carnot efficiency is:

(a) 100% **(b)** 50% **(c)** 75% **(d)** 25%

45. If a refrigerator removes 1200 J and rejects 1800 J, work input is:

(a) 1800 J **(b)** 3000 J **(c)** 1200 J **(d)** 600 J

46. If a gas absorbs 500 J and does 300 J work, ΔU is:

(a) 300 J **(b)** 200 J **(c)** 800 J **(d)** 500 J

47. Entropy change when 1 mol ice melts at 273 K with 6.01 kJ is about:

(a) 273 J K^{-1} **(b)** 6.01 J K^{-1} **(c)** 22.0 J K^{-1} **(d)** 0.022 J K^{-1}

48. A gas at 20°C heated to 80°C at constant P changes volume from 400 mL to about:

(a) 400 mL **(b)** 100 mL **(c)** 800 mL **(d)** 482 mL

49. A gas at 2 atm and 300 K heated to 450 K at constant V has pressure:

(a) 4 atm **(b)** 1.5 atm **(c)** 3 atm **(d)** 2 atm

50. Thermodynamics mostly deals with:

(a) only measurement of heat **(b)** conversion of heat to other energy forms **(c)** only temperature **(d)** only pressure

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Answer Key with Brief Reasons

1. Answer: (c) PV = nRT — General ideal gas equation.

2. Answer: (b) 1.38 × 10^{-23} J K^{-1} — $k_B = R/N_A$.

3. Answer: (a) 8.31 J mol^{-1} K^{-1} — R ≈ 8.31 J/mol·K.

4. Answer: (c) low pressure and high temperature — Molecules are far apart and fast.

5. Answer: (d) P ∝ 1/V — PV = constant.

6. Answer: (c) V ∝ T — V/T = constant.

7. Answer: (c) P ∝ T — P/T = constant.

8. Answer: (a) temperature — No intermolecular potential energy.

9. Answer: (a) Q = ΔU + W — Heat supplied = internal energy increase + work by system.

10. Answer: (d) PΔV — W = PΔV.

11. Answer: (c) work done — Area = PΔV.

12. Answer: (b) ΔU = 0 — Temperature constant means U constant.

13. Answer: (a) Q = 0 — No heat exchange.

14. Answer: (a) PV^γ = constant — γ = Cp/Cv.

15. Answer: (c) falls — Work comes from internal energy.

16. Answer: (a) slow — Heat must flow to keep T constant.

17. Answer: (b) steeper than isothermal — Pressure falls faster.

18. Answer: (a) heat into work — It operates in a cycle.

19. Answer: (c) W/Q_1 — η = work output / heat input.

20. Answer: (d) $1 - T_2/T_1$ — Temperatures in kelvin.

21. Answer: (d) Carnot engine — Carnot theorem.

22. Answer: (d) some heat must be rejected — Second law requires cold reservoir.

23. Answer: (c) a heat engine in reverse — Work moves heat from cold to hot.

24. Answer: (c) Q_2/W — COP = heat removed / work input.

25. Answer: (a) external work is supplied — Work drives the heat transfer.

26. Answer: (a) ΔQ/T — For reversible heat transfer.

27. Answer: (a) increases — Second law in entropy form.

28. Answer: (a) disorder — It measures molecular randomness.

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29. Answer: (a) maximum entropy and no available work — Energy becomes unavailable for work.

30. Answer: (c) increases — Liquid is more disordered.

31. Answer: (d) no net heat flow — Temperatures are equal.

32. Answer: (c) zero — System returns to initial state.

33. Answer: (d) negative if by-system convention — Compression work on system is negative in Q = ΔU + W_by.

34. Answer: (b) perfectly elastic — Elastic collisions conserve kinetic energy.

35. Answer: (b) molecular collisions with walls — Momentum transfer produces force.

36. Answer: (b) absolute temperature — KE_avg ∝ T.

37. Answer: (c) temperature increases — P ∝ T.

38. Answer: (b) temperature increases — V ∝ T.

39. Answer: (a) volume decreases — P ∝ 1/V.

40. Answer: (b) infinitely slow and retraceable — It can be reversed without net change.

41. Answer: (c) irreversible — It cannot be exactly retraced.

42. Answer: (a) complete conversion of heat to work in a cycle — Some heat must be rejected.

43. Answer: (b) reservoir temperatures — $\eta = 1 - T_2/T_1$.

44. Answer: (b) 50% — $1 - 300/600 = 0.5$.

45. Answer: (d) 600 J — $W = Q_1 - Q_2$.

46. Answer: (b) 200 J — $\Delta U = Q - W$.

47. Answer: (c) 22.0 J K^{-1} — $\Delta S = 6010/273$.

48. Answer: (d) 482 mL — $V_2 = 400 \times 353/293$.

49. Answer: (c) 3 atm — $P_2 = 2 \times 450/300$.

50. Answer: (b) conversion of heat to other energy forms — It studies heat, work and energy.

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